
CAPSTONE PROJECT

AUDIO SIGNAL PROCESSING ASSISTANT

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OUTLINE

- Problem Statement
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- Algorithm & Deployment
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- Future Scope
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PROBLEM STATEMENT

- Modern audio circuit designers often face challenges in debugging and optimizing circuits such as preamps, filters, and amplifiers.

Common issues include:

- ☐ Unexpected noise in audio amplifiers
- ☐ Distortion in op-amp circuits
- ☐ Ineffective grounding and shielding

Manual troubleshooting is time-consuming and requires deep domain expertise.

There is a need for an intelligent assistant that can guide users through audio signal processing challenges efficiently.

PROPOSED SOLUTION

The proposed system is an Audio Signal Processing Assistant Agent that:

- Answers user queries related to audio circuit design and debugging
- Suggests filter configurations, grounding techniques, and noise reduction strategies
- Uses a knowledge base and signal processing models to provide contextual advice
- Supports real-time interaction via a chatbot or voice interface

Key Features:

- Query understanding using NLP
- Circuit simulation insights
- Adaptive recommendations based on user input

SYSTEM APPROACH

- Platform Used:
 - IBM Cloud Lite
 - IBM watsonx Studio
- Technologies & Services:
 - Granite3-3-8b-instruct: Large language model for intelligent query handling
 - LangGraph + ReAct Architecture: For structured reasoning and dynamic response generation
 - IBM Cloud Object Storage: For storing circuit metadata and logs
 - IBM Cloud Functions: For backend logic and API handling
 - IBM Watson Assistant : For chatbot interface integration
- Libraries Required:
 - `scipy.signal`, `matplotlib`, `transformers`, `ibm_watson`, `flask`

ALGORITHM & DEPLOYMENT

Model Used:

- Granite3-3-8b-instruct (IBM watsonx)

Architecture:

- LangGraph for multi-step reasoning
- ReAct for combining reasoning and action in responses

Data Input:

- Natural language queries
- Optional circuit parameters (e.g., gain, frequency, op-amp type)

Training Process:

- Prompt engineering and instruction tuning
- Domain-specific examples for audio circuit debugging

RESULT

- Agent Capabilities:
 - Answers queries like:
 - “Why is my amplifier noisy?”
 - “How do I reduce distortion in my op-amp?”
 - “Suggest a low-pass filter for 1kHz.”
- Example Output (from Granite model):
 - Suggests checking power supply ripple, grounding layout, shielding
 - Recommends RC values for filter design
 - Provides maintenance tips for audio systems

RESULT

Why is my amplifier noisy?"

The screenshot displays the IBM watsonx interface for an 'Audio Signal Processing Assistant Agent'. The left sidebar contains a 'Build' section with the following components:

- Setup**: A dropdown menu.
- Configuration**: A section with two dropdowns: 'Framework' set to 'LangGraph' and 'Architecture' set to 'ReAct'.
- Instructions**: A text area containing the instruction: "you are An Audio Signal Processing Agent assists in designing or debugging circuits like preamps, filters, and amplifiers. It can answer queries such as 'Why is my audio amplifier producing noise?' or 'How do I reduce distortion in my op-amp circuit?'. The agent provides filter design suggestions, grounding tips, and noise elimination techniques."
- Knowledge**: A dropdown menu.
- Tools**: A section with two buttons: 'Add a tool' and 'Create custom tool'.
- Added tools (3)**: A section showing 'Google search' with a description: 'Retrieve information from the internet with the Google search engine'.

The right sidebar shows the 'Agent preview' chat interface. The chat history includes:

- User (MV)**: "Hi"
- watsonx Agent**: "Hello! How can I assist you with audio signal processing today?"
▶ How did I get this answer?
- User (MV)**: "Why is my amplifier noisy?"
- watsonx Agent**: "I'm sorry for the inconvenience. Audio amplifier noise can be caused by several factors, including poor power supply regulation, inadequate grounding, electromagnetic interference (EMI), or improper component selection. To troubleshoot, you might want to check your power supply, ensure good grounding practices, shield sensitive components, and review your circuit design for potential issues."

"For a more detailed analysis, I can perform a search for common causes of noise in amplifier circuits. Would you like me to do that?"
▶ How did I get this answer?

The bottom of the chat interface features a text input field with the placeholder 'Type something...' and a send button.

RESULT

How do I reduce distortion in my op-amp?

The screenshot displays the IBM watsonx interface for configuring and previewing an agent. The top navigation bar includes the IBM watsonx logo, an 'Upgrade' button, a help icon, a notification bell, and user account information for 'Vegi Mahidar's Account' in 'Dallas'. The main header shows the project path: 'Projects / Audio Signal Processing Assistant Agent / watsonx Agent', along with 'Share feedback', 'Autosaved 8:13 PM', and buttons for 'New agent +', 'Deploy', and a lock icon.

The left sidebar contains a 'Build' section with a dropdown menu for 'Model: granite-3-3-8b-instruct'. Below this are sections for 'Setup', 'Configuration', 'Instructions', 'Knowledge', and 'Tools'. The 'Configuration' section shows 'LangGraph' selected for the 'Framework' and 'ReAct' for the 'Architecture'. The 'Instructions' section contains a text box with the following content: 'you are An Audio Signal Processing Agent assists in designing or debugging circuits like preamps, filters, and amplifiers. It can answer queries such as "Why is my audio amplifier producing noise?" or "How do I reduce distortion in my op-amp circuit?". The agent provides filter design suggestions, grounding tips, and noise elimination techniques.' Below the instructions are buttons for 'Add a tool' and 'Create custom tool'. The 'Tools' section shows 'Added tools (3)' with a 'Google search' tool listed.

The right sidebar, titled 'Agent preview', shows a conversation log. The user's input is 'How do I reduce distortion in my op-amp?'. The watsonx Agent's response is: 'To reduce distortion in an operational amplifier (op-amp) circuit, consider the following techniques: 1. **Proper Biasing:** Ensure that the op-amp is operating within its linear region. This usually means that the output should not be driven to the supply rails. 2. **Feedback Network Design:** A well-designed feedback network can significantly reduce distortion. Make sure that the feedback components (usually resistors) are of high quality and properly matched. 3. **Frequency Compensation:** Some op-amps may exhibit increased distortion at certain frequencies. Frequency compensation techniques, such as using a dominant pole compensation, can help maintain linearity across the audio frequency range. 4. **Power Supply Decoupling:** Use bypass capacitors to decouple the power supply and reduce the impact of supply noise on the op-amp's performance. 5. **Component Selection:** High-quality, low-noise components, especially operational amplifiers, can greatly reduce distortion. For a more specific solution, I can look up strategies for reducing distortion in op-amp circuits tailored to your particular application. Would you like me to proceed with that?'. At the bottom of the preview, there is a text input field with the placeholder 'Type something...' and a send button.

RESULT

Suggest a low-pass filter for 1kHz.”

IBM watsonx Upgrade ? 🔔 Vegi Mahidar's Account ▾ Dallas ▾ VM

Projects / Audio Signal Processing Assistant Agent / watsonx Agent Share feedback β Autosaved 8:14 PM New agent + Deploy 🔗 ⓘ

Build

AI Model: granite-3-3-8b-instruct ⚙️ <

Setup

Configuration

Framework Architecture

LangGraph ▾ ReAct ▾

Instructions

[Advanced configuration](#)

you are An Audio Signal Processing Agent assists in designing or debugging circuits like preamps, filters, and amplifiers. It can answer queries such as "Why is my audio amplifier producing noise?" or "How do I reduce distortion in my op-amp circuit?". The agent provides filter design suggestions, grounding tips, and noise elimination techniques."

Knowledge

Tools

Add a tool Create custom tool

Added tools (3)

Google search ⚙️ 🗑️

Retrieve information from the internet with the Google search engine

Agent preview

MV You: 08:14 PM

Suggest a low-pass filter for 1kHz.”

PM watsonx Agent 08:14 PM

To design a low-pass filter with a cutoff frequency of 1 kHz, we can use a simple first-order RC (Resistor-Capacitor) filter. The cutoff frequency (f_c) for an RC low-pass filter is given by:

$$f_c = \frac{1}{2\pi RC}$$

where R is the resistance in ohms and C is the capacitance in farads.

Let's choose some standard values for R and C to achieve a 1 kHz cutoff. If we select $R = 1 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$, we can calculate the cutoff frequency:

$$f_c = \frac{1}{2\pi \times 1000 \text{ }\Omega \times 10 \times 10^{-6} \text{ F}} \approx 159.15 \text{ Hz}$$

This value is lower than the desired 1 kHz cutoff, so we need to adjust our component values. Let's try $R = 10 \text{ k}\Omega$ and $C = 100 \text{ }\mu\text{F}$:

$$f_c = \frac{1}{2\pi \times 10000 \text{ }\Omega \times 100 \times 10^{-6} \text{ F}} \approx 15.915 \text{ kHz}$$

This cutoff frequency is higher than our target. To hit exactly 1 kHz, we can use an online calculator or formula to find the exact values. Here's a more precise calculation using the online calculator:

For a cutoff frequency of 1 kHz with $R = 1 \text{ k}\Omega$, $C \approx 1.59 \text{ nF}$.

If you prefer using a second-order filter (like Sallen-Key topology) for a steeper roll-off, I can help you design that as well. Which would you prefer?

Type something... 🔍 ➤

CONCLUSION

The Audio Signal Processing Assistant Agent:

- Simplifies the debugging and design process for audio circuits
- Offers intelligent, context-aware suggestions
- Reduces time and effort for engineers and students
- Bridges the gap between theoretical knowledge and practical troubleshooting

FUTURE SCOPE

Planned Enhancements:

- To extend the utility of the Audio Signal Processing Assistant Agent, integration with real-time circuit simulation tools like LTspice and Multisim is planned. This will allow users to simulate and validate circuit behavior directly through the assistant, enabling more accurate diagnostics and design suggestions. Additionally, voice-based interaction will be introduced to support hands-free debugging, making the tool more accessible in lab environments where manual input may be impractical.

Scope Expansion & Offline Use:

- The assistant will also expand its domain to include RF and analog communication circuits, offering support for tasks such as impedance matching, antenna tuning, and modulation analysis. To foster creativity, generative AI will be incorporated to suggest schematic diagrams based on user intent and constraints. Furthermore, edge deployment is planned to enable offline usage, ensuring the assistant remains functional in bandwidth-limited or secure lab settings.

REFERENCES

Audio Signal Processing Agent (Project Example)	GitHub Project Repository
IBM Granite AI Models Overview	IBM Granite Official Page
Granite Model Architecture & Use Cases	IBM Developer Blog – Granite 3.2
Granite Model Background	IBM Granite – Wikipedia
IBM Cloud Lite Services	IBM Cloud Free Tier
IBM Cloud Lite Quickstart Guide	IBM Cloud Docs – Lite Plan
My Audio Signal Processing Agent Project	GitHub Repository (My Project)

IBM CERTIFICATIONS

- Getting Started With AI:



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- Journey to cloud:



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- RAG Lab:





THANK YOU